FIRST TECH CHALLENGE | TEAM #8680

KRAKEN-PINION · ENGINEERING PORTFOLIO

Season: CenterStage (2023-24) | Location: Mequon, Wisconsin, USA
Organization / School: Homestead High School

MAJOR AWARDS & JUDGING RECOGNITION

· Inspire Award Winner · FIRST World Championship (Houston, TX)
· Control Award Winner · Sponsored by Arm
· Winning Alliance Captain · World Championship

EXECUTIVE SUMMARY

FTC #8680 Kraken-Pinion achieved the highest honor in FIRST Tech Challenge by securing the World Championship Inspire Award through relentless engineering iteration, community outreach, and high-scoring autonomous reliability.

CORE SUBSYSTEMS & MECHANISMS

· Dual Compliant Roller Intake with Auto-Reverse
· Linear Slide Virtual 4-Bar Pixel Depositor
· Pocketed Low-CG Mecanum Drivetrain
· Spring-Loaded Drone Launching Mechanism

MECHANICAL & CAD ENGINEERING HIGHLIGHTS

· Dual-Roller Pixel Intake: 14 iterations of 3D-printed compliant rollers tuned to acquire pixels regardless of field debris or entry orientation.
· Virtual 4-Bar Scoring Mechanism: Linkage matched to the 60-degree backdrop slope, placing pixels with zero bounce.
· Pocketed 6061 Aluminium Chassis: CNC-machined structural rails with internal wire raceways and low-friction Delrin bearings.

AUTONOMOUS NAVIGATION & SOFTWARE

· 60Hz Visual Odometry: Limelight 3G camera fused with optical flow sensors to maintain field coordinates at high velocities.
· 98.4% Autonomous Consistency: 2+4 autonomous pixel cycle executed successfully across all World Championship division matches.
· Driver-Assist Snap-to-Backdrop: Automated alignment orienting the drivetrain parallel to AprilTags on the backdrop.

COMMUNITY OUTREACH & STEM ADVOCACY

CONNECTING & MOTIVATING INITIATIVES

· Empowering Rural STEM: Donated robotics starter parts and hosted 40+ virtual design reviews for under-resourced FIRST LEGO League and FTC teams.

· Global Engineering Office Hours: Conducted weekly open mentoring sessions supporting international teams across Europe, Asia, and the Americas.

· Public Library STEM Days: Interactive demonstrations engaging over 1,500 students and families with live drive-the-robot sessions.